An exact Bessel-function identity for the thermally averaged Doppler monopole operator

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Abstract

We prove, from first principles and with every step carried to its conclusion, that the Maxwell–Jüttner thermal average of the monopole component $\mathcal{D}_{000}(q)$ of the relativistic Doppler operator is an elementary ratio of modified Bessel functions of the second kind,

$$\langle \mathcal{D}_{000}(q) \rangle_\theta \;=\; \frac{K_{2q+3}(1/\theta) - K_1(1/\theta)}{2\,(q+1)(q+2)\,\theta\,K_2(1/\theta)},$$

valid for all $q \in \mathbb{C}$ and all dimensionless temperatures $\theta = k_B T_e/m_e c^2 > 0$. The proof rests on a single analytic input — the Schläfli–Basset integral representation of K_ν, which we prove in Appendix A by the ordinary-differential-equation method — together with elementary hyperbolic algebra. Two independent consistency checks (the pointwise normalisation $\mathcal{D}_{000}(0) \equiv 1$ and the recurrence $K_3 - K_1 = (4/z)K_2$) close the argument.

1 Statement

Throughout, $\eta \geq 0$ denotes the electron rapidity, so that its dimensionless momentum, Lorentz factor and speed are

$$p = \sinh\eta = \frac{|\boldsymbol{p}|}{m_e c}, \qquad \gamma = \cosh\eta = \sqrt{1+p^2}, \qquad \beta = \tanh\eta, \tag{1}$$

and we write $J_w(\eta) = 2\sinh(w\eta)/w$ for $w \neq 0$ (with $J_0 = 2\eta$). The monopole component of the Doppler operator — the round trip of a power-law photon spectrum ν^{-q} into the electron rest frame and back, evaluated on its own weight — has the closed form (Doppler Operator Project, `proofs/closed-forms.html`)

$$\boxed{\mathcal{D}_{000}(q;\eta) \;=\; \frac{J_{2+q}(\eta)\,J_{1+q}(\eta)}{4\,\gamma\,p^2} \;=\; \frac{\sinh\big((2+q)\eta\big)\,\sinh\big((1+q)\eta\big)}{(1+q)(2+q)\,\cosh\eta\,\sinh^2\eta}} \tag{2}$$

The second equality is immediate from $J_w = 2\sinh(w\eta)/w$ and $\gamma p^2 = \cosh\eta\sinh^2\eta$. Our object of study is the average of (2) over a thermal (Maxwell–Jüttner) ensemble of electrons.

* `adrien.cristian@gmail.com`. Companion note to the Doppler Operator Project, https://adriencr19.github.io/doppler-operator/. Every displayed identity is machine-verified in the harness `verification/38_bessel_D00.wl`.

Theorem 1 (Bessel identity for the thermal Doppler monopole)**.** *Let $\theta > 0$ and let $\langle \cdot \rangle_\theta$ denote the isotropic Maxwell–Jüttner average of Definition 1. Then for every $q \in \mathbb{C}$,*

$$\langle \mathcal{D}_{000}(q) \rangle_\theta \;=\; \frac{K_{2q+3}(1/\theta) - K_1(1/\theta)}{2\,(q+1)(q+2)\,\theta\,K_2(1/\theta)}. \tag{3}$$

The right-hand side is entire in q: the apparent poles at $q = -1$ and $q = -2$ are removable (Remark 1).

2 The Maxwell–Jüttner ensemble

Definition 1 (Maxwell–Jüttner distribution and average)**.** The relativistic (Jüttner) equilibrium distribution of electron momenta at dimensionless temperature $\theta = k_B T_e / m_e c^2$ is isotropic in $\boldsymbol{p}$ with one-dimensional momentum density

$$f_{\mathrm{MJ}}(p;\theta)\,\mathrm{d}p \;=\; \frac{p^2\,\exp\!\big(-\sqrt{1+p^2}\,/\,\theta\big)}{\theta\,K_2(1/\theta)}\,\mathrm{d}p, \qquad p \in [0,\infty), \tag{4}$$

and for any function g of the electron speed we set $\langle g \rangle_\theta = \displaystyle\int_0^\infty g(p)\,f_{\mathrm{MJ}}(p;\theta)\,\mathrm{d}p$.

That (4) is a probability density — i.e. that the normalising constant is exactly $\theta K_2(1/\theta)$ — is the content of the following proposition, proved so that no ingredient of Theorem 1 is imported unproven.

Proposition 1 (Normalisation)**.** *For every $z > 0$,*

$$\int_0^\infty p^2\,e^{-z\sqrt{1+p^2}}\,\mathrm{d}p \;=\; \int_0^\infty \sinh^2\eta\,\cosh\eta\,e^{-z\cosh\eta}\,\mathrm{d}\eta \;=\; \frac{K_2(z)}{z}. \tag{5}$$

In particular $\int_0^\infty f_{\mathrm{MJ}}(p;\theta)\,\mathrm{d}p = 1$ upon setting $z = 1/\theta$.

Proof. The substitution (1), $p = \sinh\eta$, $\mathrm{d}p = \cosh\eta\,\mathrm{d}\eta$, $\sqrt{1+p^2} = \cosh\eta$, turns the first integral into the second and shows the integrand decays like $e^{-z\cosh\eta} \sim e^{-\frac{1}{2}z\,e^\eta}$, so both integrals converge absolutely for $z > 0$. For the value, reduce the cubic in $\cosh\eta$ by the elementary identity $\cosh^3\eta = \frac{1}{4}\cosh 3\eta + \frac{3}{4}\cosh\eta$:

$$\sinh^2\eta\,\cosh\eta = (\cosh^2\eta - 1)\cosh\eta = \cosh^3\eta - \cosh\eta = \tfrac{1}{4}\big(\cosh 3\eta - \cosh\eta\big). \tag{6}$$

Hence, using the representation $K_\nu(z) = \int_0^\infty e^{-z\cosh\eta}\cosh(\nu\eta)\,\mathrm{d}\eta$ (Lemma 1) for $\nu = 3$ and $\nu = 1$,

$$\int_0^\infty \sinh^2\eta\,\cosh\eta\,e^{-z\cosh\eta}\,\mathrm{d}\eta = \tfrac{1}{4}\big(K_3(z) - K_1(z)\big) = \tfrac{1}{4}\cdot\frac{4}{z}K_2(z) = \frac{K_2(z)}{z}, \tag{7}$$

where the penultimate step is the recurrence $K_3(z) - K_1(z) = (4/z)K_2(z)$ proved in Lemma 2. $\square$

3 Proof of Theorem 1

The argument is four exact moves: change of variables, cancellation of the kinematic prefactor, a product-to-sum identity, and term-by-term application of the Bessel representation. No term is discarded and no asymptotic approximation is made.

Step 1 (change of variables). By Definition 1 and (2),

$$\langle \mathcal{D}_{000}(q)\rangle_\theta = \frac{1}{\theta K_2(1/\theta)} \int_0^\infty p^2\, e^{-\sqrt{1+p^2}/\theta}\, \mathcal{D}_{000}(q;p)\, \mathrm{d}p. \tag{8}$$

Substituting $p = \sinh\eta$ as in (1) sends the phase-space measure to $p^2\,\mathrm{d}p = \sinh^2\eta\,\cosh\eta\,\mathrm{d}\eta$ and $\sqrt{1+p^2} \mapsto \cosh\eta$, giving

$$\langle \mathcal{D}_{000}(q)\rangle_\theta = \frac{1}{\theta K_2(1/\theta)} \int_0^\infty \underbrace{\sinh^2\eta\,\cosh\eta}_{\text{measure}}\, \underbrace{\frac{\sinh\big((2+q)\eta\big)\sinh\big((1+q)\eta\big)}{(1+q)(2+q)\,\cosh\eta\,\sinh^2\eta}}_{\mathcal{D}_{000}(q;\eta)}\, e^{-\cosh\eta/\theta}\,\mathrm{d}\eta. \tag{9}$$

Step 2 (exact cancellation of the kinematic prefactor). The measure $\sinh^2\eta\,\cosh\eta$ and the denominator $\cosh\eta\,\sinh^2\eta$ of $\mathcal{D}_{000}$ are identical and cancel *exactly* — this is the structural reason a thermal average of the Doppler monopole is elementary. Equation (9) collapses to

$$\langle \mathcal{D}_{000}(q)\rangle_\theta = \frac{1}{(1+q)(2+q)\,\theta K_2(1/\theta)} \int_0^\infty \sinh\big((2+q)\eta\big)\,\sinh\big((1+q)\eta\big)\, e^{-\cosh\eta/\theta}\,\mathrm{d}\eta. \tag{10}$$

The remaining integrand is $O(e^{(2q+3)\eta})$ at worst against $e^{-\cosh\eta/\theta}$, so the integral converges absolutely for every $q \in \mathbb{C}$ and every $\theta > 0$; this justifies all manipulations below and the analytic continuation in q.

Step 3 (product-to-sum). By the addition formulae, $2\sinh A\sinh B = \cosh(A+B) - \cosh(A-B)$. With $A = (2+q)\eta$ and $B = (1+q)\eta$, so that $A+B = (2q+3)\eta$ and $A-B = \eta$,

$$\sinh\big((2+q)\eta\big)\,\sinh\big((1+q)\eta\big) = \tfrac{1}{2}\Big(\cosh\big((2q+3)\eta\big) - \cosh\eta\Big). \tag{11}$$

Step 4 (Bessel representation). Insert (11) into (10) and apply the Schläfli–Basset representation $K_\nu(z) = \int_0^\infty e^{-z\cosh\eta}\cosh(\nu\eta)\,\mathrm{d}\eta$ (Lemma 1), termwise with $z = 1/\theta$, to the orders $\nu = 2q+3$ and $\nu = 1$:

$$\begin{aligned}\int_0^\infty \sinh\big((2+q)\eta\big)\sinh\big((1+q)\eta\big)\, e^{-\cosh\eta/\theta}\,\mathrm{d}\eta &= \tfrac{1}{2}\int_0^\infty \Big(\cosh\big((2q+3)\eta\big) - \cosh\eta\Big) e^{-\cosh\eta/\theta}\,\mathrm{d}\eta \\ &= \tfrac{1}{2}\Big(K_{2q+3}(1/\theta) - K_1(1/\theta)\Big). \end{aligned}\tag{12}$$

Substituting (12) into (10) yields

$$\langle \mathcal{D}_{000}(q)\rangle_\theta = \frac{K_{2q+3}(1/\theta) - K_1(1/\theta)}{2\,(1+q)(2+q)\,\theta\,K_2(1/\theta)}, \tag{13}$$

which is (3). □

4 Consistency and physical content

Corollary 1 (Cold limit is the identity)**.** *At $q = 0$, $\langle \mathcal{D}_{000}(0)\rangle_\theta = 1$ for every $\theta > 0$.*

Proof. Two independent verifications. *(i) Pointwise.* Setting $q = 0$ in (2) and using $J_2 = 2\sinh\eta\cosh\eta$, $J_1 = 2\sinh\eta$ gives $\mathcal{D}_{000}(0;\eta) = (2\sinh\eta\cosh\eta)(2\sinh\eta)/(4\cosh\eta\sinh^2\eta) = 1$ identically, so any normalised average of it is 1. *(ii) From* (3). With $z = 1/\theta$ the numerator is $K_3(z) - K_1(z) = (4/z)K_2(z)$ by Lemma 2, and the denominator is $2\cdot 1\cdot 2\cdot z^{-1}K_2(z) = (4/z)K_2(z)$; the ratio is 1. $\square$

The agreement of (i) and (ii) is a nontrivial test: it forces the Bessel recurrence $K_3 - K_1 = (4/z)K_2$, the same identity used in Proposition 1, and thus ties the normalisation of the ensemble to the $q = 0$ value of the observable.

Remark 1 (The average is entire in q). The prefactor $[(1+q)(2+q)]^{-1}$ in (3) appears to produce simple poles at $q = -1, -2$, but the numerator vanishes there: at $q = -1$ the order is $2q+3 = 1$, so $K_{2q+3} - K_1 = K_1 - K_1 = 0$, and at $q = -2$ the order is $2q+3 = -1$, so $K_{-1} - K_1 = K_1 - K_1 = 0$ (recall $K_{-\nu} = K_\nu$). Each apparent pole is therefore a removable $0/0$, consistent with the left-hand side $\langle \mathcal{D}_{000}(q)\rangle_\theta$ being an absolutely convergent integral — hence finite — for all $q \in \mathbb{C}$ (Step 2). The identity (3) holds as an equality of entire functions of q.

Remark 2 (Nonrelativistic expansion). Expanding (3) for small θ (equivalently, averaging the momentum series $\mathcal{D}_{000}(q;p) = 1 + \frac{1}{3}q(q+3)\,p^2 + \frac{2}{45}q(q^3 + 6q^2 + 5q - 12)\,p^4 + \cdots$ against the Jüttner moments $\langle p^2\rangle = 3\theta + \frac{15}{2}\theta^2 + \cdots$, $\langle p^4\rangle = 15\theta^2 + \cdots$) gives

$$\langle \mathcal{D}_{000}(q)\rangle_\theta = 1 + q(q+3)\,\theta + \tfrac{1}{6}\,q\big(4q^3 + 24q^2 + 35q - 3\big)\,\theta^2 + O(\theta^3). \tag{14}$$

The leading correction $q(q+3)\theta$ is the familiar thermal-SZ Compton-y weight of a power law of index q; its vanishing at $q = 0$ (and at $q = -3$, the parity-conjugate weight) recovers Corollary 1.

A The Schläfli–Basset representation and its recurrence

We prove the one analytic fact used above, so the note is self-contained. For integer order this is classical (Schläfli 1873; Basset 1889); the ODE argument below covers arbitrary complex order, which we need since $2q + 3$ need not be an integer.

Lemma 1 (Integral representation of K_ν)**.** *For every $\nu \in \mathbb{C}$ and every $z > 0$,*

$$I_\nu(z) := \int_0^\infty e^{-z\cosh t}\cosh(\nu t)\,\mathrm{d}t \;=\; K_\nu(z). \tag{15}$$

Proof. Convergence and smoothness. For $t \to \infty$, $\cosh(\nu t) = O(e^{|\operatorname{Re}\nu|\,t})$ while $e^{-z\cosh t} = O\big(e^{-\frac{1}{2}ze^t}\big)$, so the integrand and all its t- and z-derivatives are dominated by an integrable function locally uniformly in $z > 0$; hence I_ν is smooth on $(0,\infty)$ and we may differentiate under the integral sign.

I_ν solves the modified Bessel equation. Write $\mathcal{L}[u] = z^2u'' + zu' - (z^2+\nu^2)u$. Differentiating (15) twice in z (so $u' = -\int \cosh t\, e^{-z\cosh t}\cosh\nu t\,\mathrm{d}t$ and $u'' = \int \cosh^2 t\, e^{-z\cosh t}\cosh\nu t\,\mathrm{d}t$) and using $z^2\cosh^2 t - z^2 = z^2\sinh^2 t$,

$$\mathcal{L}[I_\nu](z) = \int_0^\infty e^{-z\cosh t}\left(z^2\sinh^2 t - z\cosh t - \nu^2\right)\cosh(\nu t)\,\mathrm{d}t =: \int_0^\infty T(t)\,\mathrm{d}t. \tag{16}$$

The integrand is an exact t-derivative. Indeed, put

$$F(t) = e^{-z\cosh t}\big(-z\sinh t\,\cosh\nu t - \nu\sinh\nu t\big). \tag{17}$$

Differentiating and collecting terms (using $\frac{\mathrm{d}}{\mathrm{d}t}e^{-z\cosh t} = -z\sinh t\, e^{-z\cosh t}$),

$$
\begin{aligned}
e^{z\cosh t}F'(t) &= -z\sinh t\big(-z\sinh t\,\cosh\nu t - \nu\sinh\nu t\big)\\
&\qquad + \big(-z\cosh t\,\cosh\nu t - z\nu\sinh t\,\sinh\nu t - \nu^2\cosh\nu t\big)\\
&= z^2\sinh^2 t\,\cosh\nu t + \cancel{z\nu\sinh t\,\sinh\nu t} - z\cosh t\,\cosh\nu t - \cancel{z\nu\sinh t\,\sinh\nu t} - \nu^2\cosh\nu t\\
&= \big(z^2\sinh^2 t - z\cosh t - \nu^2\big)\cosh\nu t,
\end{aligned}
$$

the two $z\nu\sinh t\,\sinh\nu t$ terms cancelling exactly, so that $F'(t) = e^{-z\cosh t}\big(z^2\sinh^2 t - z\cosh t - \nu^2\big)\cosh\nu t = T(t)$. Since $F(0) = 0$ (the bracket vanishes at $t = 0$) and $F(t) \to 0$ as $t \to \infty$ (the Gaussian-in-e^t decay of $e^{-z\cosh t}$ beats the exponential growth of the bracket for $z > 0$), (16) gives $\mathcal{L}[I_\nu](z) = F(\infty) - F(0) = 0$. Thus I_ν is a solution of the modified Bessel equation of order ν.

Identification with K_ν. The solution space is spanned by I_ν (regular; growing like $e^z/\sqrt{2\pi z}$) and K_ν (recessive; decaying like $\sqrt{\pi/2z}\,e^{-z}$) as $z \to +\infty$. By Watson's lemma applied to (15) about $t = 0$, where $\cosh t = 1 + \frac{1}{2}t^2 + O(t^4)$ and $\cosh\nu t = 1 + O(t^2)$,

$$
I_\nu(z) = \int_0^\infty e^{-z(1+t^2/2)}\big(1 + O(t^2)\big)\,\mathrm{d}t = e^{-z}\sqrt{\frac{\pi}{2z}}\,\big(1 + O(z^{-1})\big) \qquad (z \to +\infty), \tag{18}
$$

independent of ν to leading order. Hence I_ν is the recessive solution and $I_\nu = c\,K_\nu$; matching the leading coefficient to $K_\nu(z) \sim \sqrt{\pi/2z}\,e^{-z}$ fixes $c = 1$. Therefore $I_\nu = K_\nu$. □

Lemma 2 (Difference recurrence)**.** *For every $\nu \in \mathbb{C}$ and $z > 0$, $K_{\nu+1}(z) - K_{\nu-1}(z) = \dfrac{2\nu}{z}\,K_\nu(z)$. In particular $K_3(z) - K_1(z) = (4/z)K_2(z)$.*

Proof. Using (15) and $\cosh(\nu+1)t - \cosh(\nu-1)t = 2\sinh\nu t\,\sinh t$,

$$
\begin{aligned}
K_{\nu+1}(z) - K_{\nu-1}(z) &= \int_0^\infty e^{-z\cosh t}\,2\sinh\nu t\,\sinh t\,\mathrm{d}t = -\frac{2}{z}\int_0^\infty \sinh\nu t\,\frac{\mathrm{d}}{\mathrm{d}t}\Big(e^{-z\cosh t}\Big)\mathrm{d}t\\
&= -\frac{2}{z}\Big[\sinh\nu t\,e^{-z\cosh t}\Big]_0^\infty + \frac{2}{z}\int_0^\infty \nu\cosh\nu t\,e^{-z\cosh t}\,\mathrm{d}t = \frac{2\nu}{z}\,K_\nu(z),
\end{aligned}
$$

the boundary term vanishing at both ends ($\sinh 0 = 0$; $e^{-z\cosh t} \to 0$). Setting $\nu = 2$ gives the stated special case. □

Verification

Every displayed equation is checked in `verification/38_bessel_D00.wl`: the prefactor cancellation of Step 2 and the product-to-sum (11) are symbolic zeros; the normalisation (5) and the representation (15) (including non-integer order $\nu = \frac{11}{2}$) match to ~ 30 digits; the assembled identity (3) matches direct quadrature of $\langle \mathcal{D}_{000}(q)\rangle_\theta$ to 2×10^{-30} across $q \in \{-\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}\}$ and $\theta \in \{0.02, 0.1\}$; and Corollary 1 is confirmed pointwise and via the recurrence.